ZOO-502 Addition 2 (7 September 2021)

CURRENT PAPER FILE (2021)

Admire group of zoologist

How to prevent alveoli expansion? 2

Alveoli are very small bubble-like structures that are inflated and deflated alternately.

The continuous inflation and deflation creates mechanical problems that might cause them to collapse. The air pressure inside the alveoli with thin walls may cause them to burst or join with other larger alveoli with lower pressure, forming larger sacs. This is known as alveolar collapse. It results in reduced surface area available for gas exchange.

Prevention of Alveolar Collapse

Alveolar collapse normally does not occur in the lung for two reasons: • surrounding tissue helps prevent overexpansion of alveoli • presence of pulmonary surfactants Role of Pulmonary Surfactants in Preventing Collapse Due to presence of surfactant on the surface of alveoli, the wall tension increases when the surface film is expanded and decreases when it is compressed. When their volume decreases (deflation), alveoli fold and the layer of surfactant between the folds becomes thick. Thick layer of surfactant reduces the surface tension. This permits easy inflation of folded alveoli. This effect of surfactants also minimizes pressure differences between large and small alveoli, thus reducing the chance of collapse.

1. Baroreceptors? 2

Baroreceptors

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The baroreceptor system is a simple and rapidly acting control mechanism that contributes to the regulation of arterial blood pressure.

Baroreceptors in Vertebrates

Baroreceptors are widely distributed in the arterial system of vertebrates.

• The central cardiovascular system of amphibia, reptiles and mammals has unmyelinated baroreceptors.

• These unmyelinated baroreceptors only respond to pressures above normal, initiating reflexes that reduce arterial blood pressure and thus protect the animal from damaging increases in blood pressure

• Mammals also have myelinated baroreceptors which respond to blood pressures below normal, thus protecting the animal from prolonged periods of reduced blood pressure.

• The myelinated baroreceptors are present in the mammalian carotid sinus, aortic arch, subclavian artery, pulmonary arteries and the wall of right atrium

2. What is hypoxia? How it effects aquatic animals? 3

Hypoxia is a condition in which the body or a region of the body is deprived of adequate oxygen supply.

Respiratory Responses of Aquatic Animals to Hypoxia

Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.

Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability. Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability.

For example, many fishes increase gill ventilation rate in hypoxic conditions. Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

3. Physiological adoptaion of ectotherms in hot environment? 3

Strategies to Live in Hot Environments

Most ectotherms actively respond to prevent elevation of their body temperatures to the critical thermal maximum. They tend to eliminate excess heat gained from the environmental by behavioral and physiological mechanisms.

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Physiological Adaptations

Most ectotherms have high heat conductance that allows them to radiate heat rapidly on moving to less warm places. Above 40oC many reptiles start panting, just like birds and mammals to increase heat loss through respiratory evaporation that causes cooling. The marine iguanas can regulate heart rate and flow of blood to the surface tissues to either absorb or eliminate heat. The Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool its internal body temperature—a process similar to sweating in mammals.

4. Role of nervous system in maintaining blood pressure?

Neural Mechanisms

Neural mechanisms respond to changes in arterial pressure or blood gas levels sensed at specific sites.

When those changes occur, the cardiovascular centers in the medulla oblongata and pons adjust cardiac output and peripheral resistance to maintain blood pressure and ensure adequate blood flow.

5. Alimentary canal?

The alimentary canal of all animals has a basic tubular organization that extends between two openings, a mouth and an anus.

Mammalian Digestive Tract

The mammalian digestive tract is a representative example of an alimentary canal.

6. Objectives of ethology?

Ethology

The scientific study of animal behavior in their natural environment is called as ethology

Objectives of Ethology

1. Understanding the stimuli that elicit behavior.
2. Understanding the physiological mechanisms that mediate the response.
3. Understanding how an animal's experience during growth and development influence the response.
4. Understanding how the behavior aids in survival and reproduction.
5. Understanding the behavior's evolutionary history

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7. Animal orientation and examples?

Animal Orientation

Orientation is the ability of an animal to determine its position in space and move predictably with respect to specific stimuli.

Types of Orienting Movements

1. Kinesis
2. taxis

Kinesis

Kinesis is a change in activity or turning rate in response to a change in location. For example, sow bugs (terrestrial crustaceans) exhibit a kinesis in response to variation in humidity. They become more active in dry areas and less active in humid areas. They do not move toward or away from specific conditions, but their increased movement under dry conditions makes it more likely that they will leave a dry area and encounter a moist area, where they survive better.

Taxis

A taxis is an oriented movement toward (positive taxis) or away (negative taxis) from some stimulus. For example, trout and many other river fishes automatically swim or orient themselves in an upstream direction (toward the current). This taxis keeps the fish from being swept away and keeps it facing the direction from which food will come. Cockroaches move away from the stimulus of light, which is called negative phototaxis. An animal that turns toward the light is said to exhibit positive phototaxis.

8. Chemoreceptors?

Chemoreceptors are receptor cells that are specialized for acquiring information about the chemical environment and transmitting it to other neurons.

Chemoreceptors can be divided into two categories:

- Gustatory (taste) receptors
- Olfactory (smell) receptors

Three types of chemoreceptors monitor changes in O_2 and CO_2 and pH levels in arterial blood:

- Peripheral Chemoreceptors
- Central Chemoreceptors

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• Type-J Chemoreceptors

9. Bile composition?

The vertebrate liver secretes bile that does not contain any digestive enzymes. However, it contains many salts which are essential for the digestion of fats.

Composition of Bile

Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile **organic salts** are manufactured by the liver from cholesterol and amino acids complexed with sodium.

The **bile pigments** are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin coming from ruptured red blood cells.

10. Gills function in fishes?

Gills

Gills are effective respiratory organs for gas exchange in water.

They have a thin, moist and highly vascularized layer of epidermis that permits gas exchange through water medium.

Gills as Respiratory Organs of Fishes

Fishes live in an environment that contains less than 2.5% of the oxygen present in air.

To maintain adequate levels of oxygen in their bloodstream, fishes must pass large quantities of water across gill surfaces and extract the small amount of oxygen present in the water.

Fishes have gills which are efficient respiratory organs for gas exchange in water.

11. Ornithine-urea cycle?

Ornithine-Urea Cycle

• All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.

• In this cycle, two ammonia and one CO_2 molecule are added to ornithine to form arginine.

• Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

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12. Steps of gas transfer?

Gas Transfer System

A gas-transfer system in animals takes in oxygen from the environment takes it to the issues through blood and brings CO_2 from tissues, transporting it to the gas exchange surface and removes it by diffusion.

Such a system involves four basic steps:

1. Breathing movements, which assure a continual supply of air or water to the respiratory surface (e.g., lungs or gills)
2. Diffusion of O_2 and CO_2 across the respiratory epithelium
3. Bulk transport of gases by the blood
4. Diffusion of O_2 and CO_2 across capillary walls between blood and mitochondria in tissue cells

13. Enzymes control secretion of pancreatic juice?

Control of Pancreatic Secretions

Pancreatic secretions are controlled by a variety of hormones including peptide hormones, gastrin, cholecystokinin, somatostatin and enkephalins.

Effect of Peptide Hormones

Acid chyme reaching the small intestine from the stomach stimulates the release of peptide hormones secretin and vasoactive intestinal peptide (VIP) by endocrine cells in the upper small intestine. These peptides stimulate secretion of pancreatic enzymes.

Effect of Gastrin

Gastrin secreted from the stomach lining also elicits a small flow of pancreatic juice in anticipation of the food coming in the duodenum.

Effect of Cholecystokinin

Secretion of pancreatic enzymes is also elicited by the upper intestinal hormone cholecystokinin that is secreted in response to fatty acids and amino acids in the intestinal chyme. This hormone also stimulates contraction of the smooth muscle wall of the gallbladder, forcing bile into the duodenum.

Effect of Somatostatin and Enkephalins

Somatostatin inhibits gastric and pancreatic secretions. Enkephalins inhibit gastric acid secretion but stimulate pancreatic enzyme secretion

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14. Vasodilator?

Local vasodilators include:

o Decreased tissue oxygen levels or increased CO2 levels.

o Lactic acid or other acids generated by tissue cells.

o Nitric oxide (NO) released from endothelial cells.

o Rising concentrations of potassium ions or hydrogen ions in the interstitial fluid.

o Elevated local temperature.

15. Receptors arteriol fibers?

Atrial Receptors The atrial walls contain many mechanoreceptive afferent fibers.

These include:

• Myelinated A-type afferent fibers

• Myelinated B-type afferent fibers

• Unmyelinated C-type afferent fibers

• Stretch-sensitive secretory cells

16. Role of hepatocytes in nutrient transfer?

In the liver, much of the glucose is taken up by hepatocytes and, under the influence of insulin, is converted into glycogen granules for storage. Only a controlled quantity is released into the circulation to be utilized by tissues.

17. Kin selection and example?

Kin Selection

Kin selection is an altruistic evolutionary strategy that favors the reproductive success of an organism's close relatives, even at a cost of organism's own survival and reproduction. Kin selection is an instance of inclusive fitness, which combines the number of offspring produced by an individual itself and the number of individuals it can ensure by supporting other close relatives (siblings).

William Hamilton proposed the idea of kin (relatives) selection to explain how selection acting on related animals can affect the fitness of an individual.

Through kin selection, a gene carried by a particular individual passes to the next generation through a related individual. The fitness of an individual is based on the genes it passes on as well as on those common genes that its relatives pass on.

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For kin selection to be effective, the individuals of a group must be able to identify their relatives. This is the reason that it is found in animals with advanced social structures i.e. primates and social insects

The altruistic kin selection behavior is most readily apparent in the act of parents sacrificing for their offspring. When parents sacrifice their own well-being to produce and aid offspring, this actually increases the fitness of the parents because it maximizes their genetic representation in the population.

18. Do vertebrates have lungs?

All vertebrate animals that live on land have lungs. When we breathe in, the muscle below the rib cage (called the diaphragm) is pulled down, and air gets sucked into the rib cage, filling the lungs

19. Stomach types in vertebrates?

Types of Stomach in Vertebrates Based on the number of chambers,

stomachs are classified as:

• Monogastric stomachs

• Digastric stomachs

Monogastric Stomachs

A monogastric stomach consists of a single strong muscular tube or sac. Carnivorous and omnivorous vertebrates have a monogastric stomach.

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe, sheep, cattle, camel, llama etc.

These stomachs have four chambers which are separated into two divisions.

• The first division consists of fermentation chambers called rumen and reticulum.

• The second division comprises the omasum and abomasum (true stomach).

20. Types of alimentary system?

Types of Alimentary Systems There are two basic plans of alimentary systems observed in animals: 1. Sac-like Gastrovascular Cavity 2. Tubular Alimentary Canal or Gu

21. Uses of crop?

Crop

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In some animals, esophagus contains a sac-like expanded section, the crop, which is used to store food before digestion.

Crop is generally found in those animals that feed infrequently.

It allows quantities of food to be stored for digestion at a later time.

Crop—Examples:

Many birds e.g. chicken. Leeches, which feed very infrequently, with weeks or months between feeding periods.

In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

22. Function and parts of colon?

Colon

The colon is about 1.5 m long and consists of four parts:

- ascending colon,
- transverse colon,
- descending colon and
- sigmoid colon.

Colon receives the undigested part of the food along with inorganic ions and excess water.

Ions and water are absorbed by the colon. As water is absorbed, indigestible wastes become increasingly solid and are converted into feces as they move along the colon by peristalsis.

It takes approximately 12-24 hours for material to travel the length of the colon and reach the rectum for elimination through the anus.

23. Aggregation?

Aggregation

• An aggregation is any form of gathering of animals or the process of coming together.

• There is minimal interaction between members of an aggregation.

• For example, several Drosophila flies living on a piece of rotting fruit.

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• A stable group of individuals of the same species that maintains a close-knit cooperative social relationship is called animal society.

• This association typically has complex social organization and extends beyond the level of mating and taking care of young.

• Such societies are observable both in invertebrates and vertebrates.

24. Core temperature?

Mammals: can maintain their core body temperature between 37°C to 38°C

. • Birds: can maintain their core body temperature closer to 40°C

25. Panting?

Panting

• Panting involves use of respiratory system to lose heat by evaporative cooling. It occurs in mammals, birds and some reptiles.

• During panting, mammals inhale through the nose and exhale through the mouth, exposing the tongue and other mouth structures to encourage further water evaporation and therefore heat loss.

• It also involves increase in breathing rate and depth (hyperventilation).

26. Systematic circuit and peripheral circuit?

The pathway followed by blood pumped from the left ventricle of the mammalian heart till its return to the right atrium is known as peripheral or systemic circulation.

Pathway of Peripheral Circulation

• The oxygenated blood is pumped by the left ventricle into the aorta which is the largest artery in the peripheral circulation.

• Branches from aorta form the arterial system and supply blood to the organs and major body regions.

• Arteries branch to form capillary beds in the tissues where oxygen is exchanged for carbon dioxide.

• The deoxygenated blood returns through the venous system to the superior and inferior vena cavae which open into the right atrium

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27. Define tubular alimentary canal?

Tubular Alimentary Canal or Gut

The complex bilateral animals have a hollow, tubular cavity which is open at both ends with two openings: mouth for ingestion and anus for egestion.

Such a tube is called a complete digestive tract or alimentary canal.

Alimentary Canal allows unidirectional flow of food materials, increasing the efficiency of digestive and absorptive processes. The food passes through the canal through various regions that have specializations for different types of digestive actions involving:

- mechanical, chemical and bacterial treatment of the food for digestion
- absorption of products of digestion
- storage of undigested waste materials

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28. Types of blood vessels?

Blood Vessels

Three types of blood vessels carry blood during circulation:

- Arteries which carry blood away from the heart to organs and tissues.
- Veins which return blood to the heart.
- Capillaries which have thin walls formed of only a single layer of endothelial cells and permit exchange of materials between blood and tissues.

29. Function of appendix?

Humans have a relatively small cecum with a finger-like extension, the appendix which is about 9 cm long.

Appendix has no role in digestion or absorption but has an undefined role in immunity as it has many lymph nodes.

1. Torpor with e.g?

Torpor

Small endotherms can not maintain their high rates of metabolic rates when they are not feeding during periods of inactivity.

During those periods, some of these animals enter a state of torpor, in which body temperature and metabolic rate decrease.

The lower basal metabolism lowers the rate of conversion of energy stores into body heat.

Thus, it is generally advantageous to allow body temperature to decrease during periods of nonfeeding and inactivity.

As the animal comes out of torpor and becomes active, it undergoes a burst of metabolic activity, through shivering or oxidation of brown-fat and its body temperature rises.

Daily torpor is practiced by many terrestrial birds.

The hummingbird is a classic example, allowing its body temperature to fall from a daytime level of about 40°C to a nighttime level as low as 13°C.

Several species of small mammals also undergo torpor (e.g., shrews)

2. Effect of gravity on atrial bp?

Effect of Gravity on Arterial blood Pressure

Gravity affects the column of blood in larger arteries. So pressure in limbs and head region varies in different positions of the body.

In lying down position, the heart is at the same level as the limbs and head with respect to gravity. So, blood pressure in the arteries of head and limbs is equal.

In a standing position, the relationship between the head, heart, and limbs changes with respect to gravity. In this position heart is below the head and about a meter above the lower limbs. So, the arterial pressure in lower limbs increases while in the head region, it decreases.

3. Cognitive maps?

Some animals guide their activity by a cognitive map, a representation in the nervous system of the spatial ,relationships between objects in an animal's surroundings. Rather than relying solely on moving from

landmark to landmark, animals using cognitive maps can navigate more flexibly and efficiently by relating landmark positions to one another.

Example

The Clark's nutcracker birds store thousands of pine seeds in fall at thousands of hiding places called mcaches, distributed over an area as large as 35 km2

. During the winter, the birds relocate many of their caches.

4. Parental care?

Parental Care

• Parental care caters the needs of the newborn and young offspring.

• It is more evolved in animals whose newborns and young cannot care for their selves.

• So, it has been an important factor in the evolution of mating systems.

5. Pro enzyme?

Proenzymes

Certain digestive enzymes, in particular proteolytic enzymes, are synthesized, stored, and released in an inactive molecular form known as a proenzyme or zymogen.

Packaging of the enzyme in an inactive form prevents self-digestion of releasing tissues/cells by the enzyme.

The proenzyme is activated by the removal of a portion of the molecule, either by the action of another enzyme specific for this purpose or through a rise in ambient acidity.

ExamplesPepsinogen-Pepsin

6. Vertebrate have lungs?

Terrestrial vertebrates (amphibians, reptiles, birds, and mammals) use a pair of lungs to exchange oxygen and carbon dioxide between their tissues and the air.

Basic Plan of Vertebrate Lung

Considerable structural variations occur among the four groups.

The generalized vertebrate lung consists of a complex network of tubes and sacs or air spaces.

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The sizes of terminal air spaces become progressively smaller in the order: amphibians, reptiles, and mammals. However the total number of air spaces per unit volume becomes greater.

Alveoli make the respiratory epithelium of lungs. Respiratory epithelium is thin and well vascularize

30. Division of oesophagus?

Esophagus

After being swallowed, food enters the esophagus which is a tube that leads from the pharynx to the stomach which is a storage as well as digestive region. Esophagus does not produce any digestive secretion or enzymes and only the previous digestive actions of saliva continue during this conduction.

31. Blubber and its roles?

Blubber is a thick layer of fat, also called adipose tissue, directly under the skin of all marine mammals. Role: As fatty tissues are little metabolically active & have little blood vascularization so it is a good insulator b/c of low thermal conductivity and prevent heat loss through body surface.

32. Difference between laminar and turbulent flow?

Laminar Flow:

Flow is streamlined and continuous. The velocity of fluid is constant at every point. Flow occurs in layers at different velocities. Flow is zero at the wall and maximal at the center, A pressure difference supplies the force required to slide adjacent layers to past each other.

Turbulent Flow:

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Direction of fluid movement is not aligned. It requires higher energy to move blood through Vessels. Turbulence also happens if smoothness is reduced by any obstruction e.g. Buildup of fatty deposits on arterial walls.

33. Fahraeu- lindqvist effect? 2

Fahraeus-Lindqvist Effect: Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahraeus Lindqvist effect. It reduces energy required to drive blood through microcirculation

34. Mechanoreceptor fibers in alveolar wall?

Ans: Atrial Receptors: Atrial walls have mechano-receptive afferent fibers including:

• Myelinated A-type afferent fibers • Myelinated B-type afferent fibers

• Unmyelinated C-type afferent fibers • Stretch-sensitive secretory cells

35. Types of vascular resistance?

Vascular Resistance "The resistance that must be overcome to push blood through the circulatory system and create flow" Types • Systemic vascular resistance (SVR) • Pulmonary vascular resistance (PVR)

36. Four accessory glands of alimentary canal?

Accessory Glands: Secrete digestive juices through ducts. The accessory glands of the mammalian digestive system are:

- Three pairs of salivary glands
- Gastric glands
- Pancreas
- Liver
- Gallbladder

37. Factors affects systemic vascular resistance /3

Factors affecting SVR

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- Elasticity of vessel wall
- Diameter of vessel
- Decreasing vessel diameter (vasoconstriction) increases SVR

  Increasing vessel diameter (vasodilation) decreases SVR

38. What is Rete Mirabile? Mention its function in the Tuna fish? 5

Ans: Rete Mirabile A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood.

Function in the Tuna fish:

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swim bladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO_2 exchangers.

39. Note on physiological adaptation of ectothermic in hot environment /5

Physiological Adaptations

• Ectotherms have high heat conductance

• Radiate heat rapidly—by moving to less warmer places

• Panting: Above 40oC many reptiles start panting—like birds and mammals

• Increases heat loss through respiratory evaporation, causing cooling.

• Regulating heart rate and flow of blood to surface tissues—to eliminate heat (e.g. marine iguanas)

• Gila monster (Helodermasuspectum) evaporates water from its cloaca to cool—a process similar to sweating in mammals

40. **Write a note on types of stomach in vertebrates /5**

Types of Stomach in Vertebrates: Based on number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs

Monogastric Stomachs

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- Consist of a single strong muscular tube or sac
- e.g. stomachs of carnivorous and omnivorous vertebrates

Digastric Stomachs

- Multichambered
- Found in ruminant mammals of order Artiodactyla
- Examples: deer, giraffe, sheep, cattle, camel, llama

Digastric Stomachs

- These stomachs have four chambers, with two divisions:
- First division consists of fermentation chambers rumen and reticulum
- Second division comprises true stomach (omasum and abomasum)

41. What is hypoxia? Responses of aquatic animals in it /5

Hypoxia • "A condition in which the body or a region of the body is deprived of adequate oxygen supply"

Hypoxia in Aquatic Environment

• O2 levels in aquatic environment vary more frequently

• Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic animals can withstand very long periods of hypoxia Responses of Aquatic Animals to Hypoxia:

• Many aquatic animals utilize anaerobic metabolic pathways to survive hypoxic periods

• Many animals adjust respiratory & cardio-vascular systems to maintain O2 absorption from hypoxic H2O

• Many fishes increase gill ventilation rate in hypoxic conditions

• Ram ventilators (e.g. tuna) widely open mouth to increase water flow over the gills

42. What are pheromones? Also describe their Role? /10

Pheromones:

• Chemical messengers released by many animals in environment to communicate. Spread through air.

Characteristics:

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• Affect the behavior of individuals of a species

• Effective at remarkably low concentrations

• Have particular odors

• Detected by olfactory receptors of receiving animal

Roles of Pheromones:

1**. Recognize Members of Species:**

Unicellular organisms recognize members of their own species. Many insects identify members of their colony.

2. **Reproductive Roles:**

Pheromones attract members of opposite sex for breeding.

Induce courtship behaviors in members of a species.

Induce spawning behavior in aquatic animals with external fertilization.

Territorial Behaviors:

Many male mammals mark their territories with pheromones.

It warns other males of their occupied area.

4. Insect's Social Order:

In a honeybee colony, pheromones produced by queen maintain hive's complex social order.

5. Alarm Calls:

Members of some species produce pheromones on getting injured.

It warns other members by the presence of danger in area.

6. Repel Predators:

Pheromones produce by the pray may repel predators.

Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

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43. Q:pristalsis

The esophagus conducts the bolus by peristaltic movement. Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls. These movements start from oral cavity and end at the distal end of rectum.

44. Q:Ornithine-Urea Cycle

• All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle

. • In this cycle, two ammonia and one CO2 molecule are added to ornithine to form arginine.

• Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

45. Q:Role of Blubber

Marine mammals (seals, walruses, sirenians, cetaceans—whales) have thick layers of insulating subcutaneous fat which is known as blubber.

Blubber is a good insulator because it has a lower thermal conductivity.

Fatty tissues are metabolically very inactive and have little blood vascularization. This prevents heat loss through body surface.

46. Q:Principle of Counter Current Exchangers

They work by:

• Regulating the extent of blood flow near the body surface

• Trapping heat within the body core.

47. Q:Mechanism of Mechanoreception

Bending or stretching of the external structure generates tension that alters the permeability of the ion channels. This change in ion permeability alters the membrane potential, resulting in a depolarization or hyperpolarization.

48. Q:Name of hormons that control pancreas.

Effect of Somatostatin and Enkephalins Somatostatin inhibits gastric and pancreatic secretions. Enkephalins inhibit gastric acid secretion but stimulate pancreatic enzyme secretion.

49. Q:three methods of surviving freezing

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• Ice-nucleating Agents (INAs) • Cryoprotectants (antifreeze substances) • Supercooling

50. Ice-nucleating Agents (INAs)

Certain animals contain ice-nucleating agents (INAs) in the extracellular fluid. INAs include many inorganic substances, microorganisms, proteins, and organic residues which can accelerate the process of crystal formation (nucleation). INAs cause the extracellular fluids to freeze more readily than the intracellular fluids.

Ice Nucleation

As ice forms in the extracellular fluid, the unfrozen extracellular fluid becomes more concentrated with solutes. This process draws water out of the cells, lowering the intracellular freezing point. In this way, animals protect their cells from making ice crystals and can withstand freezing temperature.

Example The freshwater larvae of the midge (fly) Chironomus, can survive freezing at temperatures as low as - 32°C.

Antifreeze Substances The body fluids of some cold-climate ectotherms contain cryoprotectants (antifreeze substances) e.g. glycerol, sorbitol and antifreeze proteins. These substances lower the freezing point of fluids. E.g. glycerol acting as an antifreeze can lower the freezing point of solute to as low as -17°C.

Supercooling

Ice crystals formation requires ice crystals or nuclei as seeds. The animals which can undergo supercooling keep their bodies free of ice-nucleating agents INAs in a freeze-avoidance strategy. So, their body fluids can be cooled below freezing temperature, yet remaining unfrozen because ice crystals fail to form.

Examples

• Certain Arctic bottom dwelling fishes live in a continually supercooled state and normally do not freeze unless they brush against frozen ice on the water surface. • The larvae of parasitic wasp Brachon cephi can supercool to -47°C without ice crystal formation.

Q:validilators

Vasodilators are **medications that open (dilate) blood vessels**. They affect the muscles in the walls of your arteries and veins, preventing the muscles from tightening and the walls from narrowing. As a result, blood flows more easily through your vessels.

51. Q:objective of orthology

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Understanding the stimuli that elicit behavior. 2. Understanding the physiological mechanisms that mediate the response. 3. Understanding how an animal's experience during growth and development influence the response. 4. Understanding how the behavior aids in survival and reproduction. 5. Understanding the behavior's evolutionary history.

52. Adrenal Cortex and Medulla ?

The two portions of the mammalian adrenals have different cell types, functions, and embryonic origins. The cells of the cortex are true endocrine cells, derived from non-neural, mesodermal tissue. The cells of medulla are derived from the epidermal, neural tissue during embryonic development and are functionally related to the sympathetic nervous system. Thus, like the pituitary gland, each adrenal gland is a fused endocrine and neuroendocrine gland

53. Q:types of fibre ? question incomplete

MCQ

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Structure function relationship exist at which level of biological organization

1.system level

2.cellular level

3.macromolecular level

4.all

Each molecule of can bind BBBto calcium+

20

30

40

50

Cure of which disease is not pharmacological application of neuromodulation.

Ans

1.anemia

2.narcocepy

Effort by shahram zain ud din AND Neelum Nazir..

3.dementia

4.epilepsy

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